



Indian Association for the Cultivation of Science
(Deemed to be University under *de novo* Category)
**Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's
Program/PhD Course**
End-Semester Examination-Autumn 2023

Subject: Electronics
Full Marks: 50

Subject Code(s): PHS 4104/AIS 4113/PHD 117
Time Allotted: 3 h

1. The capacitance (C) of a reverse-biased abrupt Si p^+-n (for $N_a \gg N_d$) junction has been measured as a function of the reverse bias voltage, V_r , as listed in Table -1. The pn junction cross-sectional area is $500 \mu\text{m} \times 500 \mu\text{m}$. By plotting $1/C^2$ versus V_r , obtain the built-in potential, V_0 , and the donor concentration, N_d , in the n -region. Find out the value of N_a ? (5 marks)

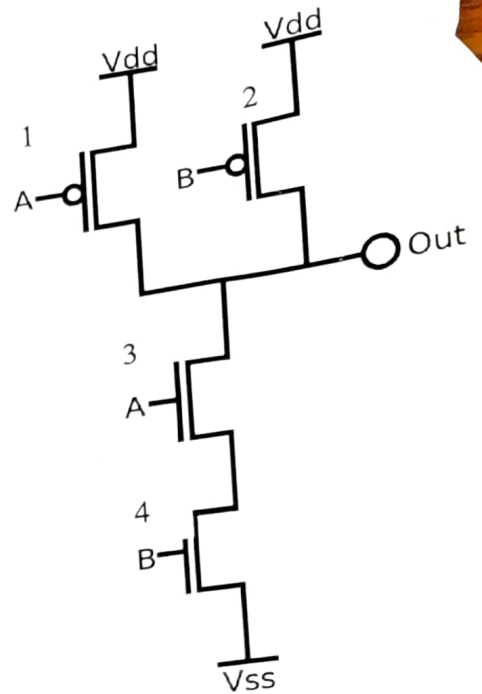
Table -1 Capacitance at various values of reverse bias (V_r)

V_r (V)	1	2	3	5	10	15	20
C (pF)	38.3	30.7	26.4	21.3	15.6	12.9	11.3

Where $\epsilon_r = 11.9$, $e = 1.602 \times 10^{-19}$ C, (built-in potential) $V_0 = 0.82$ V.

2. Show schematically the junction formation between p-type semiconductor and metal when $\phi_m < \phi_s$ where ϕ_m and ϕ_s are work functions for metal and p-type semiconductor, χ is electron affinity and E_g is band gap of p-type semiconductor. (1 marks)
- Determine whether the junction Schottky or Ohmic (1 marks)
 - Derive the expression for barrier height in equilibrium condition. (1 marks)
 - Show schematically band diagram in forward bias and reverse bias. (2 marks)
3. What is a tunnel diode, draw band bending diagram for a tunnel diode at various applied biases and explain the principle and mechanism of negative differential resistance in the tunnel diode via plotting I-V characteristics? (5 marks)
4. Derive the expression for built-in potential (V_0) of P-N junction diode in terms of donor, acceptor concentration and intrinsic carrier concentration? (5 marks)
5. A solar cell at room temperature is under an illumination of 500 W.m^{-2} and has a short circuit current, I_{sc} , of 150 mA and an open circuit voltage, V_{oc} , of 0.53 V. What are the short circuit current and the open circuit voltage when the light intensity is doubled? What are the values when the intensity is halved? (5 marks)

6. a) Write down the truth table for the following circuit and the corresponding Boolean expression.
1 & 2 $\rightarrow$ PMOS, 3 & 4 $\rightarrow$ NMOS

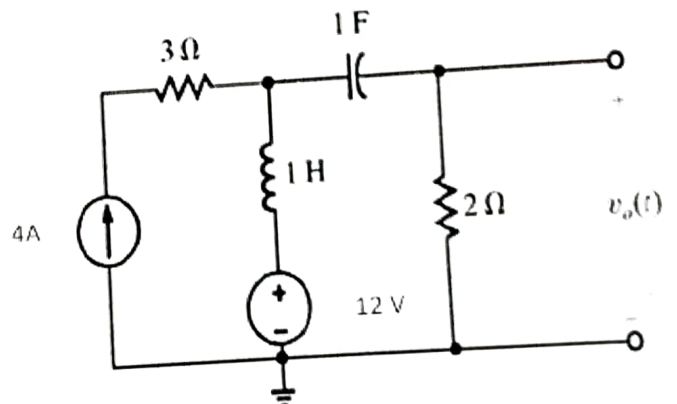


- b) An amplifier has a voltage gain of 500 without feedback. If negative feedback is applied, the gain is reduced to 100. Calculate the fraction of the output feedback. If due to ageing of components, gain without feedback falls by 20%, calculate the percentage fall in gain with feedback. Comment on the result.
- c) If an IC chip contains 1mm^2 total area of 1.2nm thin SiO_2 which leaks 10^3 A/cm^2 , will you prefer it to be in your cell phone? Give the reason.

[3+3+1]

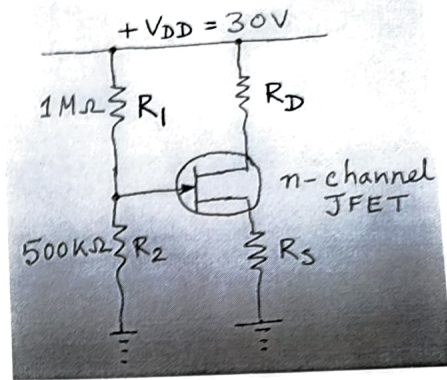
7. A zero bias time series voltage signal $v(t)$ is recorded across an unknown resistance at room temperature and has a noise level of $3.015 \times 10^{-18} \text{ V}^2 \text{ Hz}^{-1}$. What is the value of the resistance in Ohm. If the above measurement is carried out in liquid Nitrogen (77K) and at 100°C , calculate the corresponding noise levels. [3]

8. a) Draw the s-domain equivalent circuit given in the adjacent figure.
b) Find $V_0(s)$ (s-domain equivalent of $v_0(t)$) using circuit analysis. [2+8]



9. In an n-channel JFET biased by potential divider method, it is desired to set the operating point at $I_D = 2.5 \text{ mA}$ and $V_{DS} = 8 \text{ V}$. If $V_{DD} = 30 \text{ V}$, $R_1 = 1 \text{ M}\Omega$ and $R_2 = 500 \text{ k}\Omega$, find the value of R_S . The parameters of JFET are $I_{DSS} = 10 \text{ mA}$ and $V_{GS}(\text{off}) = -5 \text{ V}$.

[5]





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Master's/Integrated Master's-PhD Program/Integrated Bachelor's-Master's Program/PhD Course
Mid-Semester Examination-Autumn 2023

Subject: Classical Mechanics

Subject Code(s): PHS 4101/AIS 4110

Full Marks: 25

Time Allotted: 2 h

Answer Question No. (1) and anyone from the rest.

Q1. (a) State the fundamental lemma of variational Calculus.

(2)

(b) Deduce a condition for extremization of a function of N variables under a holonomic constraint involving those N variables.

(4)

(c) Using the method of Lagrange's multiplier, extremize the function $f(x, y) = 2 - x^2 - 2y^2$ under the constraint $g(x, y) = x^2 + y^2 = 1$.

(4)

(d) State Hamilton's Principle. Deduce Lagrange's equations of motion for an unconstrained mechanical system using Hamilton's principle.

(1+4)

Q2. (a) State and prove Noether's Theorem (you may, but need not consider a time-independent system).

(2 + 4)

(b) For a system with Lagrangian with no explicit time-dependence, prove that

$$E = \left(\sum_{i=1}^n \frac{\partial L}{\partial \dot{q}_i} \dot{q}_i \right) - L$$

is a conserved quantity (here $\dot{q}_i$ are generalized velocities).

(4)

Q3. (a) Find the conserved quantity for a particle moving in a three-dimensional space in a potential with cylindrical symmetry about a given (z , say) axis due to the symmetry.

(3)

(b) Find the expression of the geodesic (the curve of the shortest length between two points on a surface) on a three-dimensional sphere.

(4)

(c) A cube is constrained to stay on the floor with one given face down (i.e., no point on that face ever leaves the floor). Now two such constrained cubes are taken, and the center of one vertical face of a cube is joined to the center of a vertical face of the other with an extensible but non-bendable spring (the spring always remains perpendicular to both the faces it joins). Find the number of degrees of freedom of the system.

(3)



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End-Semester Examination-Autumn 2023

Subject: Classical Mechanics
Full Marks: 50

Subject Code(s): PHS 4101/AIS 4110
Time Allotted: 3 h

Answer Q1 and any three from the rest

Q1. (a) Deduce Hamilton's equations of motion for a system that obeys Euler-Lagrange equations of motion (either starting from Lagrange's equation of motion in the configuration space or formulating the variational problem in the phase-space). Mention the necessary condition under which a Hamiltonian is obtainable from a Lagrangian. (4+1)

(b) A particle is subjected to a time-dependent external influence such that its motion is described by the following Lagrangian

$$L = ap^2 + bq^3 + \alpha t$$

apq

Where a , b and α are independent of time t (p and q bear their usual meaning). Find the rate of change of energy of the system with time. (2)

Deduce the relation between the time derivatives of H and L you have used above. (4)

(c) Consider two points **A** and **B**, with **A** higher above the ground than **B** but not on the same vertical line. In what shape should one build a frictionless roller coaster track so that a car released from point **A** will reach point **B** in an extremal time? Just establish the Euler-Lagrange equation for the curve that gives the shape of the track. (3)

(d) Show that adding a total time-derivative of a function of the generalized coordinates and time to a Lagrangian produces another valid Lagrangian. (3)

(e) Three point-masses reside (while in equilibrium) on the vertices of a triangle, whose sides are massless harmonic springs with spring constant k (each mass is connected to its two adjacent masses by the springs). If we assume that the system always resides on a

horizontal plane, explain the number of zero modes the system will have. Explain if the number would change if we allow the entire system to move in three-dimensional space? (2+1)

Q2. (a) Consider a disc rolling without slipping on a flat two-dimensional floor. Show that the constraints of rolling without slipping is not integrable. (4)

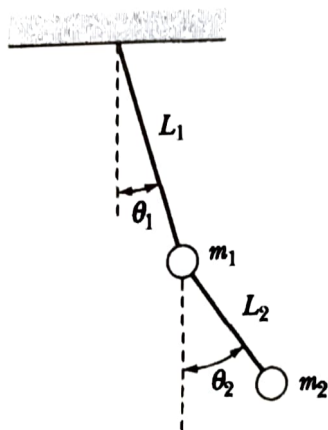
(b) Deduce Euler-Lagrange equation of motion for a system with N degrees of freedom subjected to k independent holonomic constraints. (5)

(c) What are semi-holonomic constraints? (1)

Q3. (a) Consider a close system of N point masses connected to each other by springs, with the interaction represented by a potential V involving only the cartesian coordinates of the system. Assuming that the system is executing small oscillations about its equilibrium configuration, find the Euler-Lagrange equation for the system. (4)

(b) Find the set of solutions of the equation of motion where each member of the set represents a pure oscillation (real frequency) of each particle with same frequency. (3)

(c) Consider a double pendulum – see the Figure below.



Find the Normal frequencies of the system under the special condition of having lengths of both the pendulums same ($L_1 = L_2 = \ell$) and both the masses same ($m_1 = m_2 = m$ each). (3)

Q4. Consider a Hamiltonian

$$H(p, q) = \frac{p^2}{2m}$$

and the transformation

$$P = p \cos q$$

$$Q = p \sin q$$

(a) Determine whether the equation of motion retains the Canonical form for the above Hamiltonian under the above transformation. (3)

(b) Show if the transformation is Canonical. (3)

(c) (i) Prove that a point transformation (transformation involving only the generalized coordinates and time) is always Canonical. (ii) Find a generator for the identity transformation. (2+2)

Q5. (a) Define a Symplectic matrix. Show that the Jacobian of a Canonical Transformation is a Symplectic matrix. (1 + 4)

(b) Show that Canonical Transformations form a Group. (2)

(c) Consider the following canonical transformation $(p, q) \rightarrow (P, Q)$ given by

$$\begin{aligned} Q &= q \cos \alpha - p \sin \alpha \\ P &= q \sin \alpha + p \cos \alpha \end{aligned}$$

Find the generator of type-1 for this transformation. (3)

Q6. (a) Define Poisson bracket between two observables $A(p, q)$ and $B(p, q)$ where (p, q) denotes a set of pairs of canonically conjugate generalized coordinates and momenta. Deduce the condition under which an observable $O(p, q; t)$ remains conserved under the dynamics governed by a Hamiltonian $H(p, q; t)$. (1+4)

(b) Show that if the fundamental Poisson brackets remain invariant under a transformation, then show that any Poisson bracket will remain invariant under that transformation. (3+2)

[Hint: First show that the transformation is a C.T., then show ...]

Q7. (a) Consider an infinitesimal Canonical transformation $(p, q) \rightarrow (p + \delta p, q + \delta q)$ in a $2N$ dimensional phase space, generated due to an infinitesimal change $\delta\lambda$ in a continuous parameter λ (that parametrizes the continuous canonical transformation). Show that if $G(q, p, t)$ is the generator of this transformation, then

$$\begin{aligned} \delta q_i &= \delta\lambda \frac{\partial G}{\partial P_i} \\ \delta p_i &= -\delta\lambda \frac{\partial G}{\partial q_i} \end{aligned} \quad (4)$$

(b) Show that Hamiltonian is the generator of infinitesimal time-translation.

(3)

(c) Let δO is the change in some observable $O(\{p, q\})$ due to Show that

$$\delta O = \delta \lambda \{O, G\} + \frac{\partial O}{\partial t} \delta t$$

(3)

Q8. (a) Consider the Hamiltonian of a one-dimensional simple harmonic oscillator and deduce the equation for the generator of a canonical transformation that renders all the coordinates cyclic.

(5)

(b) Solve the equation and obtain the transformation, and hence the solution for the equation of motion.

(5)

Best of Luck!